INTERNSHIP PROJECT REPORT

Role of Artificial Intelligence in Autonomous Robots

Submitted by:

KEVIN VICTOR

Internship Program

CodeAlpha Robotics & Automation

Date of Submission

April 23, 2026

Department of Robotics & Automation

Abstract

Autonomous robots represent one of the most transformative technological advancements of the twenty-first century, fundamentally reshaping the boundaries between human capability and machine intelligence. The integration of Artificial Intelligence (AI) into robotic systems has enabled machines to perceive complex environments, reason under uncertainty, learn from experience, and execute sophisticated tasks without continuous human intervention. This report provides a comprehensive examination of the pivotal role that AI plays in enabling and enhancing autonomous robotic systems.

The report covers the fundamental architecture of autonomous robots, the key AI technologies that power them—including machine learning, computer vision, decision-making algorithms, and path planning—and their deployment across diverse real-world domains such as autonomous vehicles, industrial automation, healthcare, and logistics. System-level design considerations, including the sensor-processing-decision-action pipeline, are discussed in conceptual detail. Furthermore, the report addresses the significant advantages offered by AI-driven robotics, the persistent challenges such as safety, cost, and ethical concerns, and the promising future trajectory involving AI-IoT convergence, smart city integration, and human-robot collaboration.

This report is prepared as part of the internship submission for CodeAlpha Robotics & Automation and reflects a synthesis of current research, engineering principles, and practical application insights in the domain of intelligent autonomous systems.

1. Introduction

1.1 What Are Autonomous Robots?

An autonomous robot is a robotic system capable of performing tasks and making decisions in dynamic, unstructured environments without direct human control. Unlike traditional industrial robots that operate on pre-programmed, repetitive sequences in controlled settings, autonomous robots possess the ability to sense their surroundings, process information, adapt to changing conditions, and take goal-directed actions in real time. This operational independence is what distinguishes autonomy from mere automation.

The spectrum of autonomy ranges from semi-autonomous systems—which require periodic human oversight or intervention—to fully autonomous systems that operate entirely independently across extended missions. Modern autonomous robots integrate a range of hardware subsystems including actuators, proprioceptive and exteroceptive sensors, power management units, and embedded computing platforms, all governed by sophisticated software architectures. The convergence of mechanical engineering, electrical engineering, and computer science forms the foundation upon which autonomous robotics is built.

Autonomous robots are broadly classified based on their operational domain: ground-based robots (including wheeled and legged platforms), aerial drones and unmanned aerial vehicles (UAVs), underwater autonomous vehicles (AUVs), and space exploration robots. Each category presents unique environmental constraints and design requirements, yet all share the common imperative of intelligent, independent operation.

1.2 Why Is AI Important in Robotics?

The importance of Artificial Intelligence in robotics cannot be overstated. Classical robotics relied heavily on hand-crafted rules, deterministic control algorithms, and structured environments. While effective in highly controlled settings such as assembly lines with fixed workpieces, these approaches fail catastrophically when confronted with the variability, noise, and unpredictability inherent in real-world environments.

AI introduces the capacity for generalization, perception, and adaptive learning that classical methods lack. Through machine learning, a robot can improve its performance over time based on experience rather than relying solely on a programmer's explicit instructions. Computer vision, powered by deep learning, allows robots to interpret rich visual data from cameras and depth sensors, enabling them to recognize objects, map spaces, detect obstacles, and understand human gestures. Natural language processing enables voice-controlled interactions, while reinforcement learning enables robots to discover optimal behaviors through trial-and-error interaction with their environment.

Beyond perception, AI equips robots with the reasoning capabilities necessary to plan actions under uncertainty, manage competing objectives, and respond to unanticipated events. In essence, AI transforms a robot from a mechanically precise but cognitively limited machine into an intelligent agent capable of purposeful behavior in the complexity of the real world. This transformation is at the heart of the modern autonomous robotics revolution.

2. Overview of Autonomous Robots

Autonomous robots are deployed across an extensive range of industries and applications, each leveraging AI capabilities tailored to specific operational requirements. The following subsections outline the major categories of autonomous robots currently in use or active development.

2.1 Industrial Robots

Industrial autonomous robots operate within manufacturing, warehousing, and logistics environments, performing tasks such as material handling, precision assembly, quality inspection, and collaborative human-robot production. Modern industrial robots have evolved significantly from the rigid, cage-enclosed manipulators of the 1970s. Contemporary platforms—often referred to as cobots (collaborative robots)—incorporate force-torque sensing, vision systems, and AI-driven planning to work safely alongside human operators. Companies such as FANUC, ABB, and KUKA deploy AI-enhanced robots that can adapt to part variations, self-correct errors, and optimize cycle times dynamically.

2.2 Unmanned Aerial Vehicles (Drones)

Autonomous drones represent a rapidly growing category of aerial robotic systems capable of GPS-independent navigation, obstacle avoidance, and mission-level autonomy. Their applications span aerial photography, agricultural monitoring, infrastructure inspection, search-and-rescue operations, and package delivery. The integration of AI-based visual odometry, simultaneous localization and mapping (SLAM), and deep learning-based target detection has dramatically extended the operational envelope of commercial and military drone systems. Multi-drone swarm intelligence, where coordinated fleets of UAVs collaboratively accomplish complex tasks, is an active area of research with significant implications for disaster response and defense.

2.3 Self-Driving Vehicles

Autonomous ground vehicles, particularly self-driving cars and trucks, represent perhaps the most societally impactful application of AI in robotics. These systems must navigate complex traffic environments, comply with traffic regulations, predict the behavior of other agents (pedestrians, cyclists, vehicles), and handle edge cases with the reliability expected of safety-critical systems. Companies such as Tesla, Waymo, Cruise, and Mobileye are at the forefront of deploying advanced driver assistance systems (ADAS) and fully autonomous driving capabilities. The technical complexity of autonomous driving has catalyzed substantial advances in sensor fusion, real-time inference on embedded hardware, and safety-critical AI system design.

2.4 Service Robots

Service robots are designed to interact with humans and perform tasks in social, domestic, or professional settings. This category includes medical robots (surgical assistants, rehabilitation robots, and hospital logistics robots), hospitality robots (hotel receptionists and restaurant servers), educational robots, and personal care robots for elderly or disabled individuals. Service robots must exhibit a high degree of social intelligence, including the ability to interpret human intent, respond to verbal and non-verbal cues, and operate safely in unpredictable human-populated environments. AI-driven natural language understanding and human-robot interaction models are central to their design.

3. Role of Artificial Intelligence in Autonomous Robots

Artificial Intelligence underpins the core cognitive functions of autonomous robots. The following sections detail the principal AI technologies and their specific contributions to robotic autonomy.

3.1 Machine Learning

Machine learning (ML) provides robots with the ability to acquire knowledge and improve performance from data and experience, rather than from explicit programming. Supervised learning algorithms enable robots to learn sensory-motor mappings, object classification, and predictive maintenance models from labeled training datasets. Unsupervised learning allows robots to discover structure in unlabeled environmental data, facilitating tasks such as terrain classification and anomaly detection.

Reinforcement learning (RL) is particularly powerful for robotic control, enabling agents to learn policies that maximize cumulative reward through interaction with a simulated or real environment. Deep reinforcement learning—combining neural networks with RL—has produced remarkable results in robot locomotion, dexterous manipulation, and game-playing. Transfer learning reduces the data and computational requirements of deploying ML models to new environments by leveraging knowledge acquired in related domains, a capability of particular value in robotics where real-world data collection is often expensive and time-consuming.

3.2 Computer Vision

Computer vision constitutes one of the most critical perceptual capabilities of autonomous robots, enabling them to extract meaningful information from visual data captured by cameras, LiDAR systems, and depth sensors. Convolutional neural networks (CNNs) have achieved human-level performance on image classification benchmarks and form the backbone of modern robotic vision pipelines.

Key computer vision tasks in autonomous robotics include object detection and recognition (identifying and localizing objects of interest within a scene), semantic segmentation (assigning class labels to every pixel of an image), 3D reconstruction (building volumetric models of the environment from 2D imagery or point cloud data), visual odometry (estimating the robot's motion by tracking visual features across sequential frames), and

SLAM (simultaneously building a map of the environment and localizing the robot within it). Advanced vision transformers and multi-modal architectures that fuse visual data with other sensor modalities are pushing the frontier of robotic perception.

3.3 Decision Making

Autonomous robots must continuously make decisions ranging from low-level reflex responses (withdrawing from an obstacle) to high-level strategic planning (determining the most efficient route to accomplish a mission). AI provides several frameworks for robotic decision-making under uncertainty and complexity.

Probabilistic graphical models such as Bayesian networks and Markov decision processes (MDPs) offer principled mathematical frameworks for reasoning about uncertain states and outcomes. Behavior trees and hierarchical finite state machines provide modular, interpretable architectures for organizing robot behavior across multiple levels of abstraction. In recent years, large language model (LLM)-based reasoning systems have been explored as high-level planners for robotic agents, enabling robots to interpret natural language instructions and translate them into executable action sequences. Multi-agent decision-making frameworks address scenarios where multiple robots must coordinate their actions to achieve shared objectives.

3.4 Path Planning

Path planning is the problem of determining a collision-free trajectory from a starting configuration to a goal configuration, subject to the kinematic and dynamic constraints of the robot and the geometry of the environment. Classical path planning algorithms such as A*, Dijkstra's algorithm, and the Rapidly-exploring Random Tree (RRT) family have been widely deployed in robotics and remain relevant in many applications.

AI-enhanced path planning introduces additional capabilities beyond classical methods. Deep learning-based planners can generalize across diverse environments without requiring explicit map representations. Motion prediction models anticipate the future trajectories of dynamic obstacles such as pedestrians and vehicles, enabling proactive rather than purely reactive navigation. Hierarchical planning architectures decompose complex long-horizon tasks into manageable sub-problems, each solved by specialized planners. In multi-robot systems, distributed path planning algorithms enable coordinated navigation without centralized computation, improving scalability and fault tolerance.

4. System Architecture of Autonomous Robots

The functional architecture of an autonomous robot can be conceptually decomposed into four tightly integrated processing stages: Sensing, Processing, Decision-Making, and Action Execution. This pipeline forms a closed-loop control cycle that continuously repeats during autonomous operation.

4.1 Stage 1: Sensing

The sensing stage constitutes the robot's interface with its physical environment. A comprehensive sensor suite typically includes exteroceptive sensors—which measure properties of the external environment—and proprioceptive sensors, which provide data on the robot's internal state. Common exteroceptive sensors include RGB cameras for visual scene capture, LiDAR (Light Detection and Ranging) units for high-resolution 3D spatial mapping, radar sensors for robust distance measurement in adverse weather conditions, ultrasonic sensors for proximity detection, and infrared sensors for thermal imaging. Proprioceptive sensors such as inertial measurement units (IMUs), wheel encoders, joint position sensors, and force-torque sensors provide data on the robot's velocity, orientation, joint angles, and contact forces.

Raw sensor data is inherently noisy, incomplete, and subject to calibration errors. Sensor fusion algorithms—such as Extended Kalman Filters (EKF) or particle filters—combine data from multiple heterogeneous sensors to produce statistically robust estimates of the robot's state and the state of its environment. This fused perception layer forms the foundation upon which all subsequent processing and decision-making is built.

4.2 Stage 2: Processing and Perception

The processing stage involves extracting actionable information from the raw fused sensor data. This encompasses environment modeling (constructing and maintaining an internal representation of the robot's surroundings), object detection and tracking (identifying, classifying, and estimating the motion of relevant objects), scene understanding (interpreting the semantic meaning of the environment), and state estimation (determining the robot's own position and orientation within the world frame, typically via SLAM). AI inference engines—often implemented on specialized embedded hardware such as GPUs, TPUs, or neuromorphic processors—execute these computationally intensive tasks at the speeds required for real-time operation.

4.3 Stage 3: Decision-Making and Planning

Given an internal model of the world, the decision-making stage determines what the robot should do next. This stage operates at multiple temporal and spatial scales: at the strategic level, task planners determine the sequence of high-level actions required to accomplish the mission objective; at the tactical level, motion planners generate collision-free trajectories through the environment; at the reactive level, real-time controllers modulate low-level control signals to track the planned trajectory in the presence of disturbances. Uncertainty is explicitly modeled at this stage, with probabilistic planners accounting for sensor noise, actuation errors, and unpredictable agent behavior.

4.4 Stage 4: Action Execution

The action execution stage translates planning outputs into physical commands applied to the robot's actuators: electric motors, hydraulic systems, pneumatic actuators, or end-effectors. Low-level control algorithms such as PID controllers, model predictive controllers (MPC), or learning-based inverse dynamics models ensure that the robot's physical motion closely tracks the desired trajectory. Feedback from proprioceptive sensors closes the control loop, enabling the system to detect and compensate for deviations caused by disturbances, friction, and modeling errors. The resulting physical actions—locomotion, manipulation, communication, or environmental modification—constitute the robot's observable output and produce changes in the environment that will be measured by sensors in the next sensing cycle.

5. Real-World Applications

5.1 Tesla — Autonomous Driving

Tesla's Autopilot and Full Self-Driving (FSD) systems represent one of the most ambitious commercial deployments of AI in autonomous vehicles. Tesla's approach is distinguished by its heavy reliance on a vision-only sensor architecture, eschewing LiDAR in favor of a multi-camera system processed by a custom AI accelerator chip, the Tesla Dojo training supercomputer, and a neural network inference stack deployed across its global fleet. The FSD system employs occupancy network models that generate real-time 3D representations of the environment, trajectory prediction networks that model the likely motion of surrounding agents, and end-to-end learned driving policies trained on petabytes of real-world driving data collected from millions of production vehicles. Tesla's fleet-learning approach enables rapid improvement of the AI system through continuous collection of edge-case data, illustrating the power of large-scale real-world data collection in advancing autonomous driving capabilities.

5.2 Amazon — Warehouse Robotics

Amazon's fulfillment centers deploy one of the largest fleets of autonomous robots in the world, comprising Kiva/Amazon Robotics drive units, robotic arms for picking and stowing, and autonomous mobile robots for inventory management. The Kiva robots autonomously navigate warehouse floors, transporting entire inventory pods to stationary human pickers, dramatically reducing the distance workers must travel per shift. Amazon's Robin and Sparrow robotic systems utilize AI-powered computer vision to identify and handle individual items from mixed bins, a task that requires sophisticated grasp planning and object recognition. Amazon's AI-driven warehouse systems collectively process millions of items daily, demonstrating the operational scale achievable through AI-enabled robotic automation.

5.3 Boston Dynamics

Boston Dynamics has become synonymous with cutting-edge dynamic robotic locomotion and manipulation. Their humanoid robot Atlas employs model predictive control and learned balance policies to execute acrobatic maneuvers—including backflips, parkour sequences, and dynamic jumping—that were previously considered beyond the reach of robotic systems. The Spot quadruped robot leverages AI-based perception and terrain-adaptive gait control to navigate unstructured environments including construction sites, offshore oil platforms, and disaster zones, performing inspection and data collection tasks autonomously. Boston Dynamics' research illustrates how AI-driven control can enable robots to operate in the highly variable, dynamically challenging environments that are pervasive in the physical world.

5.4 Healthcare Robots

In the healthcare domain, AI-powered robots are transforming surgical procedures, patient care, and hospital logistics. The da Vinci Surgical System employs AI-assisted image guidance and robotic manipulation to enable minimally invasive surgery with precision exceeding human manual dexterity, reducing patient recovery times and complication rates. Autonomous medication dispensing and delivery robots navigate hospital corridors to deliver pharmaceuticals and laboratory specimens, reducing nursing workload and medication errors. Rehabilitation robots equipped with AI-driven adaptive control adjust therapeutic exercise parameters in real time based on patient performance data, optimizing recovery trajectories. Socially assistive robots are deployed in memory care facilities to provide cognitive stimulation and companionship to patients with dementia, improving quality of life outcomes.

6. Advantages of AI-Driven Autonomous Robots

6.1 Operational Efficiency

AI-powered autonomous robots can operate continuously without the fatigue, attention lapses, or shift-change interruptions inherent in human labor. In logistics and manufacturing environments, this translates directly to higher throughput, faster cycle times, and improved on-time delivery performance. Path optimization algorithms minimize travel distances and energy consumption within facilities. Predictive maintenance AI models monitor robot health in real time, scheduling maintenance interventions before failures occur and thereby maximizing operational uptime. The cumulative effect of these efficiency gains can reduce operational costs significantly while simultaneously increasing output volume.

6.2 Precision and Accuracy

Robotic systems executing AI-guided tasks achieve levels of positional precision and repeatability that consistently surpass human manual performance. In surgical robotics, AI-guided instruments can achieve sub-millimeter positioning accuracy, enabling procedures that would be technically infeasible without robotic assistance. In semiconductor manufacturing, robotic wafer handling systems operate within tolerances measured in micrometers, a degree of precision physically impossible for human operators to maintain over extended periods. AI-driven quality inspection systems using high-resolution computer vision detect surface defects, dimensional deviations, and assembly errors at rates and reliabilities that exceed human visual inspection, dramatically reducing defect escape rates.

6.3 Automation of Hazardous and Repetitive Tasks

Autonomous robots are uniquely suited to tasks that are dangerous, physically demanding, or monotonously repetitive for human workers. In nuclear decommissioning, autonomous robots can enter environments with radiation levels lethal to humans, performing decontamination and structural assessment tasks remotely. In mining and underground construction, autonomous drill rigs and haulage vehicles operate in environments subject to rockfall, toxic gas exposure, and confined space hazards. In disaster response, autonomous drones and ground vehicles can enter collapsed structures or chemically contaminated areas to locate survivors and assess conditions before human responders are deployed. The substitution of robots for humans in such hazardous roles represents a direct reduction in workplace injury and fatality rates.

7. Challenges and Limitations

7.1 Safety and Reliability

Safety is the paramount challenge in the deployment of autonomous robots, particularly in environments shared with humans. AI systems—especially those based on deep learning—are vulnerable to adversarial inputs, distributional shift (degraded performance when operating in conditions that differ from the training distribution), and rare but catastrophic failure modes. Ensuring that autonomous robots behave safely across the full distribution of possible real-world conditions, including low-probability edge cases, requires extensive validation and testing strategies that are fundamentally more complex than those applicable to deterministic systems. Formal verification of AI-based control systems remains an open research problem. Regulatory frameworks for certifying autonomous systems in safety-critical applications—particularly autonomous vehicles and medical robots—are still evolving, creating uncertainty for developers and operators.

7.2 High Development and Deployment Costs

The development of autonomous robotic systems requires substantial investment in research and development, custom hardware, sensor integration, AI model training infrastructure, and software engineering. High-precision sensors such as solid-state LiDAR units, force-torque sensors, and stereo camera systems represent significant per-unit hardware costs. The computational infrastructure required to train large AI models—thousands of GPUs operating for weeks or months—constitutes a barrier accessible only to well-funded organizations. Furthermore, deployment of autonomous systems in regulated industries such as healthcare and transportation requires extensive certification processes that add time and cost. These economic barriers limit the adoption of advanced autonomous robotics to large enterprises and well-capitalized startups, slowing diffusion across smaller industrial and service sector operators.

7.3 Ethical and Societal Concerns

The widespread deployment of AI-powered autonomous robots raises profound ethical questions that the engineering community must engage with seriously. The displacement of human workers by robotic automation generates legitimate concerns about employment, economic inequality, and the social fabric of communities whose industries are transformed by automation. Autonomous weapons systems—military robots capable of selecting and engaging targets without human authorization—raise urgent questions about accountability, proportionality, and the ethics of delegating lethal decisions to machines. Data privacy concerns arise from the large-scale environmental sensing performed by deployed robot fleets. Questions of liability—who bears legal responsibility when an autonomous robot causes harm?—remain inadequately resolved in most legal jurisdictions. Algorithmic bias in AI perception systems can lead to discriminatory outcomes if systems perform differently across demographic groups.

8. Future Scope

8.1 AI and Internet of Things (IoT) Integration

The convergence of AI-powered autonomous robots with the Internet of Things (IoT) represents one of the most promising directions in next-generation robotic systems. IoT-enabled environments—in which physical objects, infrastructure, and sensors are interconnected and continuously generating data—provide autonomous robots with a dramatically enriched situational awareness that extends far beyond their onboard sensors. A robot operating in an IoT-enabled smart factory can receive real-time data from production line sensors, inventory management systems, logistics tracking platforms, and environmental monitoring devices, enabling it to make decisions with a comprehensive, contextually rich understanding of the broader operational environment. Edge-cloud computing architectures will enable robots to leverage distributed AI inference across local embedded processors and cloud computing resources, optimizing the trade-off between latency and computational capability.

8.2 Smart Cities and Infrastructure

The concept of the smart city—urban environments instrumented with pervasive sensing, data analytics, and automated management systems—provides a transformative context for the deployment of autonomous robots at scale. Autonomous delivery robots and drones will reshape last-mile logistics within urban environments, reducing traffic congestion and delivery costs. Autonomous inspection drones will continuously monitor the structural health of bridges, tunnels, and utility infrastructure, enabling proactive maintenance before failures occur. Self-driving public transit vehicles will improve mobility for elderly and disabled citizens while reducing urban transport emissions. Robotic systems will play a central role in urban waste management, environmental monitoring, and emergency response, contributing to cities that are simultaneously more efficient, more sustainable, and more resilient.

8.3 Human-Robot Collaboration

The future of autonomous robotics is not one of wholesale human replacement but rather one of deep, fluid collaboration between humans and robots, each contributing their complementary strengths to shared tasks. Next-generation cobots will leverage AI models of human intent, emotion, and attention to anticipate their human partners' needs and adapt their behavior accordingly, achieving levels of collaborative fluency that current systems cannot approach. Augmented reality interfaces will enable intuitive communication between human operators and robot teammates, while shared autonomy frameworks will allow humans and robots to fluidly exchange control authority based on situational demands. Advances in explainable AI (XAI) will make robotic decision-making more transparent and interpretable to human collaborators and supervisors, building the trust necessary for effective human-robot teams in high-stakes environments.

9. Conclusion

This report has presented a comprehensive examination of the role of Artificial Intelligence in enabling and advancing autonomous robotic systems. From the foundational sensing-processing-decision-action architecture that governs all autonomous robots, to the sophisticated AI technologies—machine learning, computer vision, decision-making frameworks, and path planning algorithms—that animate their intelligence, it is evident that AI is not merely an ancillary component of modern robotics but its defining cognitive core.

Real-world deployments by organizations such as Tesla, Amazon, Boston Dynamics, and leading healthcare institutions demonstrate that AI-powered autonomous robots are no longer a research aspiration but an operational reality, delivering measurable benefits in efficiency, precision, safety, and capability across diverse industries. At the same time, the challenges of safety assurance, economic accessibility, and ethical governance demand sustained attention from engineers, policymakers, and society at large if the benefits of autonomous robotics are to be realized equitably and responsibly.

The future trajectory of autonomous robotics—toward deeper AI-IoT integration, smart city deployment, and nuanced human-robot collaboration—promises to further amplify the transformative impact of these systems. As AI algorithms grow more capable, hardware platforms become more accessible, and regulatory frameworks mature, the scope and sophistication of autonomous robotic applications will continue to expand. The engineers and researchers entering this field today inherit both a remarkable technical opportunity and a profound responsibility to ensure that the development of autonomous intelligent machines serves the long-term benefit of humanity.

References

The following references were consulted in the preparation of this report, formatted in IEEE citation style:

[1] S. Thrun, W. Burgard, and D. Fox, Probabilistic Robotics. Cambridge, MA: MIT Press, 2005.

[2] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA: MIT Press, 2016. [Online]. Available: https://www.deeplearningbook.org

[3] B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, Robotics: Modelling, Planning and Control. London, UK: Springer, 2009.

[4] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," Nature, vol. 521, no. 7553, pp. 436-444, May 2015, doi: 10.1038/nature14539.

[5] V. Mnih et al., "Human-level control through deep reinforcement learning," Nature, vol. 518, no. 7540, pp. 529-533, Feb. 2015, doi: 10.1038/nature14236.

[6] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, "Vision meets robotics: The KITTI dataset," Int. J. Robot. Res., vol. 32, no. 11, pp. 1231-1237, Sep. 2013, doi: 10.1177/0278364913491297.

[7] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed. Cambridge, MA: MIT Press, 2018. [Online]. Available: http://incompleteideas.net/book/the-book.html

[8] Tesla, Inc., "Tesla Autopilot and Full Self-Driving Capability," Tesla Technical Documentation, 2024. [Online]. Available: https://www.tesla.com/autopilot